

INTER-PROCESS COMMUNICATION (IPC): A process is a running program. Sometimes two or more processes need to communicate or share data

Topic :

- Pipes
- Named Pipes (FIFO)

Pipes : A pipe is an IPC mechanism used to transfer data from one process to another. It usually provides one-way communication and works like a temporary data channel.

- Usually used between related processes, such as a parent and child process.
- Data is read in the same order in which it is written (FIFO order).
- A normal/anonymous pipe generally exists only while the related processes are running

Example:

A parent process creates a pipe and sends the message "Hello Child". The child process reads the message from the pipe

Parent → Pipe → Child

Named Pipes (FIFO): A Named Pipe, also called a FIFO (First In, First Out), is a pipe with a name in the file system. It allows processes that are not necessarily related to communicate with each other

- Has a name/path, so independent processes can access it.
- Data is normally read in FIFO order.
- Useful when two unrelated programs need a simple communication channel

Example:

Program A writes "Good Morning" into a FIFO. Program B opens the same FIFO and reads the message

Linux example: mkfifo mypipe

Writer: echo "Hello" > mypipe

Reader: cat mypipe